“The Urban Revolution”

Town Planning Review (1950)

V. Gordon Childe

EDITORS’ INTRODUCTION

The study of the earliest cities belongs to the fields of prehistory, anthropology, and archaeology, and more is being learned every day about the first emergence of urban civilization. V. Gordon Childe (1892–1957) is arguably the single most influential archaeologist of the twentieth century. Born in Australia, Childe won a scholarship to Queen’s College, Oxford, returned to Australia where he briefly pursued a career in left-wing politics, then returned to the UK as Professor of Archaeology at the University of Edinburgh and, later, Director of the Institute of Archaeology at the University of London.

Childe’s most important book, the one that revolutionized the world of archaeological research by laying out an entirely new theoretical framework for understanding the phases of human development throughout history and prehistory, was *Man Makes Himself* (1936). In that pioneering work, Childe threw out the “three age system” (Stone Age, Bronze Age, Iron Age) that had been left over from nineteenth-century conceptions of human historical development. In its place he proposed a series of four stages (Paleolithic, Neolithic, urban, industrial) punctuated by three “revolutions” or fundamental shifts in cultural development.

According to Childe, the first revolution – from old Stone Age hunter-gatherer cultures to settled agriculture – was the Neolithic Revolution. The second – the movement from Neolithic agriculture to complex, hierarchical systems of city-based manufacturing and trade that began during the fourth and third millennia BCE – was the Urban Revolution. And the third major shift in the record of human cultural and historical development – the only truly new development since the rise of cities – was the Industrial Revolution of the eighteenth and nineteenth centuries. It is important to bear in mind that Childe lived before the computer age and developed his typology before current debates about how information technology is revolutionizing urban society. For example, Manuel Castells (p. 229) and others consider “the informational city” operating within the “space of electronic flows” to be a radically different type of city and “the rise of the network society” to be a transformation as profound as the earlier revolutions that Childe identified.

Childe is best known for his writings on the first cities, which arose in Mesopotamia (present-day Iraq) beginning about 4000 BCE. These cities sprang up in the area bounded by the Tigris and Euphrates Rivers – often referred to as part of “the Fertile Crescent.” Plate 2, “View of Ancient Babylon,” illustrates the form these first cities took: monumental gates, massive mud-brick walls, courtyards, residences for priest-kings, and a ziggurat.

Childe’s work continues to figure prominently in ongoing debates about when, where, and why the first cities arose and in the antecedent debate about what a city is. Not everyone has accepted Childe’s notion that the shift from Neolithic to urban was a total break with the past. Evidence of ancient earthworks, wells, irrigation systems, and even continental trade networks have been traced back as far as 10,000 years in a number of areas in both the Old World and the New. Archaeologist James Mellaart has argued that evidence from the great Neolithic communities of Çatal Hüyük and Hacilar in ancient Turkey, which predate the earliest Mesopotamian cities by some thousands of years, calls the entire Childe theory into question. Some have argued that the stage of human development called "civilization" begins with Neolithic settled agriculture, not with cities. Others, like Jane Jacobs (p. 149) in *The Economy of Cities* (1969) has hypothesized an iconic urban settlement fancifully called "New Obsidian" that pre-dated – indeed invented – agriculture, fundamentally reversing the order of Childe's progression from rural to urban. Although it is almost certain that the early cities, with their massive irrigation systems and complex organization processes, radically improved the practice of Neolithic agriculture, it is still generally accepted that in most locations agriculture predated the rise of the first cities by many millennia and that the full elaboration of those cultural institutions we associate with urban life – for example, writing and record keeping and the rise of the complex apparatus of the state – only emerged with the rise of cities in Mesopotamia and elsewhere.

Not everyone agrees with Childe's definition of a city. More recent archaeologists excavating older, smaller, less culturally advanced settlements than the Mesopotamian cities Childe studied often argue that these settlements were urban enough to qualify as true cities. Scholars working in South and Central America point out that many of the cultural features Childe believed essential to the definition of a city (including the wheel, writing, and the plow) did not exist in large and culturally advanced Amerindian settlements that appear truly urban in other respects.

This new line of thinking is perhaps best explored in Peter J. Taylor, *Extraordinary Cities* (2013) with its call for a "post-Childe understanding of early cities," greater respect for the creative potential of Neolithic populations, and a new theory of early cities that does not rely on specific places of origin but focuses instead on the nature of urban processes such as "communication potential."

In the selection from *Town Planning Review* reprinted here, Childe details the constituent elements of the Urban Revolution that accompanied the initial rise of complex civilizations in Mesopotamia and elsewhere in the ancient Near East. Childe felt that the major factors motivating the transformation were rooted in the material base of the society: its means of production and its available physical and technological resources. Thus, the economic division of labor, the elaboration of socio-political hierarchies, and even the emergence of basic religious and intellectual patterns of thought characteristic of urban civilizations all rested on the underlying need to increase food production through massive irrigation systems and to protect the communities themselves through the erection of massive walls and fortifications.

Many modern scholars question the deterministic Marxist categories Childe employed. Although he stresses the importance of writing as an element of any truly urban society, Childe has been faulted for his apparent disregard of the primacy of non-material aspects of culture. His system has very little room for what Lewis Mumford (p. 110) called "the urban drama" or what Jane Jacobs (p. 149) called the "street ballet." Still, no one has ever called Childe's vision limited or ideologically cramped. On the contrary, he provided an expansive macro-historical foundation upon which generations of others have built.

A tireless researcher and writer, Childe produced a veritable stream of books, many of which are still classics. Among the most notable are: *The Dawn of European Civilization* (London: Routledge and Kegan Paul, 1925), *The Most Ancient East* (New York: Grove Press, 1928), *What Happened in History* (Harmondsworth: Penguin, 1942), *Social Evolution* (London: Watts, 1951). Other books on Mesopotamian cities include Nicholas Postgate and J.N. Postgate, *Early Mesopotamia: Society and Economy at the Dawn of History* (London and New York: Routledge, 1994), Georges Roux, *Ancient Iraq* (New York: Penguin, 1993), and C. Leonard Wooley's classics *The Sumerians* (Oxford: Oxford University Press, 1928) and *Ur of the Chaldees* (Oxford: Oxford University Press, 1929). For more recent views, see Gwendolyn Leick, *Mesopotamia: The Invention of the City* (London: Penguin, 2003) and Paul Kriwaczek, *Mesopotamia and the Birth of Civilization* (New York: Thomas Dunne, 2012). For more on Childe and his contributions to the field, consult Sally Green, *Prehistorian: A Biography of V. Gordon Childe* (Dana Point, CA: Moonraker Publications, 1981) and David R. Harris, *The Archaeology of V. Gordon Childe: Contemporary Perspectives*, 2nd edn (Chicago: University of Chicago Press, 1994).

As noted above, books of special interest include Jane Jacobs, *The Economy of Cities* (New York: Random House, 1969) and Peter J. Taylor, *Extraordinary Cities: Millennia of Moral Syndromes, World-Systems and City/State Relations* (Cheltenham: Edward Elgar, 2013). For surveys of recent research into cities in the ancient world, see Gwendolyn Leick, *Mesopotamia* and Charles Gates, *Ancient Cities: The Archaeology of Urban Life* *in the Ancient Near East and Egypt, Greece, and Rome* (London and New York: Routledge, 2003). Of special interest is Joyce Marcus and Jeremy Sabloff, *The Ancient City: New Perspectives on Urbanism in the Old and New World* (Santa Fe, NM: School for Advanced Research Press, 2009).

Earlier studies on the rise of the earliest cities elsewhere in the world that are still of interest include Mortimer Wheeler, *Civilizations of the Indus Valley and Beyond* (London: Thames & Hudson, 1966), Karl Wittfogel, *Oriental Despotism* (New Haven: Yale University Press, 1957), Basil Davidson, *The Lost Cities of Africa* (Boston: Little, Brown, 1959), Richard E.W. Adams, *Prehistoric Mesoamerica* (Norman, OK: University of Oklahoma Press, 1991), Sylvanus G. Morely and George W. Brainerd, *The Ancient Maya* (Stanford: Stanford University Press, 1956), Jacques Soustelle, *The Daily Life of the Aztecs* (New York: Macmillan, 1962), James Mellaart, *Earliest Civilizations of the Near East* (New York: McGraw-Hill, 1965) and *Catal Hüyük* (New York: McGraw-Hill, 1967), and Paul Wheatley, *The Pivot of the Four Quarters: A Preliminary Inquiry into the Origins and Character of the Ancient Chinese City* (Chicago: Aldine, 1971).

The concept of 'city' is notoriously hard to define. The aim of the present essay is to present the city historically – or rather prehistorically – as the resultant and symbol of a 'revolution' that initiated a new economic stage in the evolution of society. The word 'revolution' must not of course be taken as denoting a sudden violent catastrophe; it is here used for the culmination of a progressive change in the economic structure and social organization of communities that caused, or was accompanied by, a dramatic increase in the population affected – an increase that would appear as an obvious bend in the population graph were vital statistics available. Just such a bend is observable at the time of the Industrial Revolution in England. Though not demonstrable statistically, comparable changes of direction must have occurred at two earlier points in the demographic history of Britain and other regions. Though perhaps less sharp and less durable, these too should indicate equally revolutionary changes in economy. They may then be regarded likewise as marking transitions between stages in economic and social development.

Sociologists and ethnographers last century classified existing pre-industrial societies in a hierarchy of three evolutionary stages, denominated respectively 'savagery,' 'barbarism' and 'civilization.' If they be defined by suitably selected criteria, the logical hierarchy of stages can be transformed into a temporal sequence of ages, proved archaeologically to follow one another in the same order wherever they occur. Savagery and barbarism are conveniently recognized and appropriately defined by the methods adopted for procuring food. Savages live exclusively on wild food obtained by collecting, hunting or fishing. Barbarians on the contrary at least supplement these natural resources by cultivating edible plants and – in the Old World north of the Tropics – also by breeding animals for food.

Throughout the Pleistocene Period – the Palaeolithic Age of archaeologists – all known human societies were savage in the foregoing sense, and a few savage tribes have survived in out of the way parts to the present day. In the archaeological record barbarism began less than ten thousand years ago with the Neolithic Age of archaeologists. It thus represents a later, as well as a higher stage, than savagery. Civilization cannot be defined in quite such simple terms. Etymologically the word is connected with 'city,' and sure enough life in cities begins with this stage. But 'city' is itself ambiguous so archaeologists like to use 'writing' as a criterion of civilization; it should be easily recognizable and proves to be a reliable index to more profound characters. Note, however, that, because a people is said to be civilized or literate, it does not follow that all its members can read and write, nor that they all lived in cities. Now there is no recorded instance of a community of savages civilizing themselves, adopting urban life or inventing a script. Wherever cities have been built, villages of preliterate farmers existed previously (save perhaps where an already civilized people have colonized uninhabited tracts). So civilization, wherever and whenever it arose, succeeded barbarism.

We have seen that a revolution as here defined should be reflected in the population statistics. In the case of the Urban Revolution the increase was mainly accounted for by the multiplication of the numbers of persons living together, i.e., in a single built-up area. The first cities represented settlement units of hitherto unprecedented size. Of course it was not just their size that constituted their distinctive character. We shall find that by modern standards they appeared ridiculously small and we might meet agglomerations of population today to which the name city would have to be refused. Yet a certain size of settlement and density of population is an essential feature of civilization.

Now the density of population is determined by the food supply which in turn is limited by natural resources, the techniques for their exploitation and the means of transport and food-preservation available. The last factors have proved to be variables in the course of human history, and the technique of obtaining food has already been used to distinguish the consecutive stages termed savagery and barbarism. Under the gathering economy of savagery population was always exceedingly sparse. In aboriginal America the carrying capacity of normal unimproved land seems to have been from .05 to .10 per square mile. Only under exceptionally favourable conditions did the fishing tribes of the Northwest Pacific coast attain densities of over one human to the square mile. As far as we can guess from the extant remains, population densities in Palaeolithic and pre-neolithic Europe were less than the normal American. Moreover such hunters and collectors usually live in small roving bands. At best several bands may come together for quite brief periods on ceremonial occasions such as the Australian corroborees. Only in exceptionally favoured regions can fishing tribes establish anything like villages. Some settlements on the Pacific coasts comprised thirty or so substantial and durable houses, accommodating groups of several hundred persons. But even these villages were only occupied during the winter; for the rest of the year their inhabitants disposed in smaller groups. Nothing comparable has been found in pre-neolithic times in the Old World.

The Neolithic Revolution certainly allowed an expansion of population and enormously increased the carrying capacity of suitable land. On the Pacific Islands neolithic societies today attain a density of 30 or more persons to the square mile. In pre-Columbian North America, however, where the land is not obviously restricted by surrounding seas, the maximum density recorded is just under 2 to the square mile.

Neolithic farmers could of course, and certainly did, live together in permanent villages, though, owing to the extravagant rural economy generally practised, unless the crops were watered by irrigation, the villages had to be shifted at least every twenty years. But on the whole the growth of population was not reflected so much in the enlargement of the settlement unit as in a multiplication of settlements. In ethnography neolithic villages can boast only a few hundred inhabitants (a couple of 'pueblos' in New Mexico house over a thousand, but perhaps they cannot be regarded as neolithic). In prehistoric Europe the largest neolithic village yet known, Barkaer in Jutland, comprised 52 small, one-roomed dwellings, but 16 to 30 houses was a more normal figure; so the average, local group in neolithic times would average 200 to 400 members.

These low figures are of course the result of technical limitations. In the absence of wheeled vehicles and roads for the transport of bulky crops men had to live within easy walking distance of their cultivations. At the same time the normal rural economy of the Neolithic Age, what is now termed slash-and-burn or *jhumming*, condemns much more than half the arable land to lie fallow so that large areas were required. As soon as the population of a settlement rose above the numbers that could be supported from the accessible land, the excess had to hive off and found a new settlement.

The Neolithic Revolution had other consequences beside increasing the population, and their exploitation might in the end help to provide for the surplus increase. The new economy allowed, and indeed required, the farmer to produce every year more food than was needed to keep him and his family alive. In other words it made possible the regular production of a social surplus. Owing to the low efficiency of neolithic technique, the surplus produced was insignificant at first, but it could be increased till it demanded a reorganization of society.

Now in any Stone Age society, palaeolithic or neolithic, savage or barbarian, everybody can at least in theory make at home the few indispensable tools, the modest cloths and the simple ornaments everyone requires. But every member of the local community, not disqualified by age, must contribute actively to the communal food supply by personally collecting, hunting, fishing, gardening or herding. As long as this holds good, there can be no full-time specialists, no persons nor class of persons who depend for their livelihood on food produced by others and secured in exchange for material or immaterial goods or services.

We find indeed today among Stone Age barbarians and even savages expert craftsmen (for instance flint-knappers among the Ona of Tierra del Fuego), men who claim to be experts in magic, and even chiefs. In Palaeolithic Europe too there is some evidence for magicians and indications of chieftainship in pre-neolithic times. But on closer observation we discover that today these experts are not full-time specialists. The Ona flintworker must spend most of his time hunting; he only adds to his diet and his prestige by making arrowheads for clients who reward him with presents. Similarly a pre-Columbian chief, though entitled to customary gifts and services from his followers, must still personally lead hunting and fishing expeditions and indeed could only maintain his authority by his industry and prowess in these pursuits. The same holds good of barbarian societies that are still in the neolithic stage, like the Polynesians where industry in gardening takes the place of prowess in hunting. The reason is that there simply will not be enough food to go round unless every member of the group contributes to the supply. The social surplus is not big enough to feed idle mouths.

Social division of labour, save those rudiments imposed by age and sex, is thus impossible. On the contrary community of employment, the common absorption in obtaining food by similar devices guarantees a certain solidarity to the group. For co-operation is essential to secure food and shelter and for defence against foes, human and subhuman. This identity of economic interests and pursuits is echoed and magnified by identity of language, custom and belief; rigid conformity is enforced as effectively as industry in the common quest for food. But conformity and industrious co-operation need no State organization to maintain them. The local group usually consists either of a single clan (persons who believe themselves descended from a common ancestor or who have earned a mystical claim to such descent by ceremonial adoption) or a group of clans related by habitual intermarriage. And the sentiment of kinship is reinforced or supplemented by common rites focused on some ancestral shrine or sacred place. Archaeology can provide no evidence for kinship organization, but shrines occupied the central place in preliterate villages in Mesopotamia, and the long barrow, a collective tomb that overlooks the presumed site of most neolithic villages in Britain, may well have been also the ancestral shrine on which converged the emotions and ceremonial activities of the villagers below. However, the solidarity thus idealized and concretely symbolized, is really based on the same principles as that of a pack of wolves or a herd of sheep; Durkheim has called it 'mechanical.'

Now among some advanced barbarians (for instance tattooers or woodcarvers among the Maori) still technologically neolithic we find expert craftsmen tending towards the status of full-time professionals, but only at the cost of breaking away from the local community. If no single village can produce a surplus large enough to feed a full-time specialist all the year round, each should produce enough to keep him a week or so. By going round from village to village an expert might thus live entirely from his craft. Such itinerants will lose their membership of the sedentary kinship group. They may in the end form an analogous organization of their own – a craft clan, which, if it remain hereditary, may become a caste, or, if it recruit its members mainly by adoption (apprenticeship throughout Antiquity and the Middle Ages was just temporary adoption), may turn into a guild. But such specialists by emancipation from kinship ties, have also forfeited the protection of the kinship organization which alone under barbarism, guaranteed to its members security of person and property. Society must be reorganized to accommodate and protect them.

In pre-history specialization of labour presumably began with similar itinerant experts. Archaeological proof is hardly to be expected, but in ethnography metal-workers are nearly always full-time specialists. And in Europe at the beginning of the Bronze Age metal seems to have been worked and purveyed by perambulating smiths who seem to have functioned like tinkers and other itinerants of much more recent times. Though there is no such positive evidence, the same probably happened in Asia at the beginning of metallurgy. There must of course have been in addition other specialist craftsmen whom, as the Polynesian example warns us, archaeologists could not recognize because they worked in perishable materials. One result of the Urban Revolution will be to rescue such specialists from nomadism and to guarantee them security in a new social organization.

About 5,000 years ago irrigation cultivation (combined with stockbreeding and fishing) in the valleys of the Nile, the Tigris-Euphrates and the Indus had begun to yield a social surplus, large enough to support a number of resident specialists who were themselves released from food-production. Water-transport, supplemented in Mesopotamia and the Indus valley by wheeled vehicles and even in Egypt by pack animals, made it easy to gather food stuffs at a few centres. At the same time dependence on river water for the irrigation of the crops restricted the cultivable areas while the necessity of canalizing the waters and protecting habitations against annual floods encouraged the aggregation of population. Thus arose the first cities – units of settlement ten times as great as any known neolithic village. It can be argued that all cities in the old world are offshoots of those of Egypt, Mesopotamia and the Indus basin. So the latter need not be taken into account if a minimum definition of civilization is to be inferred from a comparison of its independent manifestations.

But some three millennia later cities arose in Central America, and it is impossible to prove that the Mayas owed anything directly to the urban civilizations of the Old World. Their achievements must therefore be taken into account in our comparison, and their inclusion seriously complicates the task of defining the essential preconditions for the Urban Revolution. In the Old World the rural economy which yielded the surplus was based on the cultivation of cereals combined with stock-breeding. But this economy had been made more efficient as a result of the adoption of irrigation (allowing cultivation without prolonged fallow periods) and of important inventions and discoveries – metallurgy, the plough, the sailing boat and the wheel. None of these devices was known to the Maya; they bred no animals for milk or meat; though they cultivated the cereal maize, they used the same sort of slash-and-burn method as neolithic farmers in prehistoric Europe or in the Pacific Islands

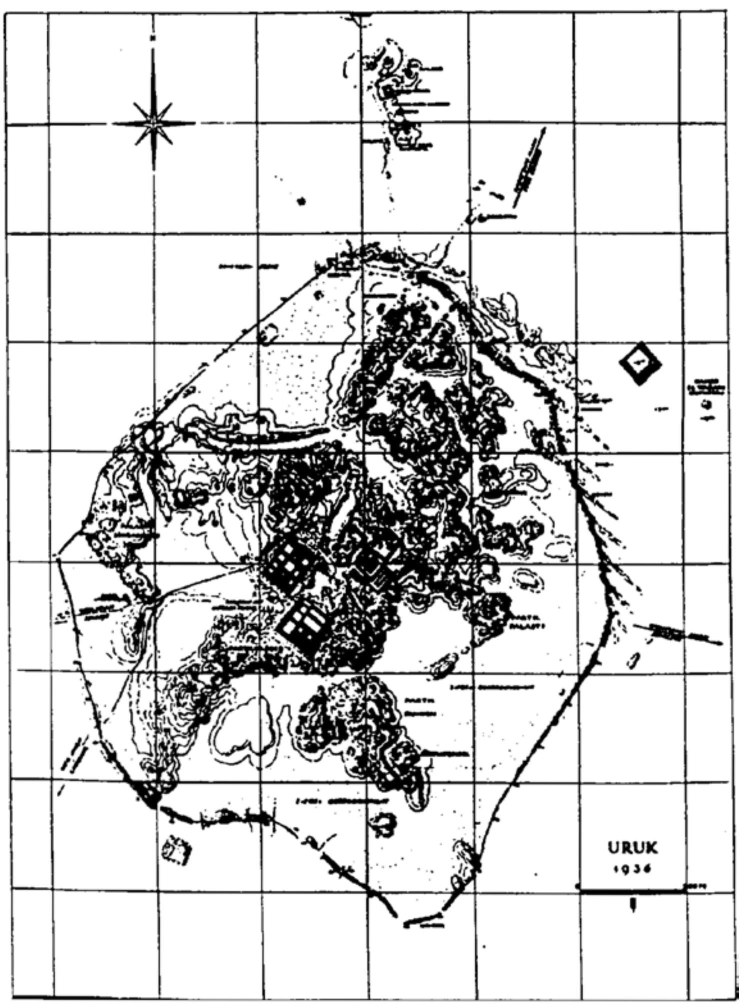

Figure 1 Plan of the city of Erek (Uruk)

today. Hence the minimum definition of a city, the greatest factor common to the Old World and the New will be substantially reduced and impoverished by the inclusion of the Maya. Nevertheless ten rather abstract criteria, all deducible from archaeological data, serve to distinguish even the earliest cities from any older or contemporary village.

(1) In point of size the first cities must have been more extensive and more densely populated than any previous settlements, although considerably smaller than many villages today. It is indeed only in Mesopotamia and India that the first urban populations can be estimated with any confidence or precision. There excavation has been sufficiently extensive and intensive to reveal both the total area and the density of building in sample quarters and in both respects has disclosed significant agreement with the less industrialized Oriental cities today. The population of Sumerian cities, thus calculated, ranged between 7,000 and 20,000; Harappa and Mohenjo-daro in the Indus valley must have approximated to the higher figure. We can only infer that Egyptian and Maya cities were of comparable magnitude from the scale of public works, presumably executed by urban populations.

(2) In composition and function the urban population already differed from that of any village. Very likely indeed most citizens were still also peasants, harvesting the lands and waters adjacent to the city. But all cities must have accommodated in addition classes who did not themselves procure their own food by agriculture, stock-breeding, fishing or collecting – full-time specialist craftsmen, transport workers, merchants, officials and priests. All these were of course supported by the surplus produced by the peasants living in the city and in dependent villages, but they did not secure their share directly by exchanging their products or services for grains or fish with individual peasants.

(3) Each primary producer paid over the tiny surplus he could wring from the soil with his still very limited technical equipment as tithe or tax to an imaginary deity or a divine king who thus concentrated the surplus. Without this concentration, owing to the low productivity of the rural economy, no effective capital would have been available.

(4) Truly monumental public buildings not only distinguish each known city from any village but also symbolize the concentration of the social surplus. Every Sumerian city was from the first dominated by one or more stately temples, centrally situated on a brick platform raised above the surrounding dwellings and usually connected with an artificial mountain, the staged tower or ziggurat. But attached to the temples were workshops and magazines, and an important appurtenance of each principal temple was a great granary. Harappa, in the Indus basin, was dominated by an artificial citadel, girt with a massive rampart of kiln-baked bricks, containing presumably a palace and immediately overlooking an enormous granary and the barracks of artisans. No early temples nor palaces have been excavated in Egypt, but the whole Nile valley was dominated by the gigantic tombs of the divine pharaohs while royal granaries are attested from the literary record. Finally the Maya cities are known almost exclusively from the temples and pyramids of sculptured stone round which they grew up.

Hence in Sumer the social surplus was first effectively concentrated in the hands of a god and stored in his granary. That was probably true in Central America while in Egypt the pharaoh (king) was himself a god. But of course the imaginary deities were served by quite real priests who, besides celebrating elaborate and often sanguinary rites in their honour, administered their divine masters' earthly estates. In Sumer indeed the god very soon, if not even before the revolution, shared his wealth and power with a mortal vice-regent, the 'City-King,' who acted as civil ruler and leader in war. The divine pharaoh was naturally assisted by a whole hierarchy of officials.

(5) All those not engaged in food-production were of course supported in the first instance by the surplus accumulated in temple or royal granaries and were thus dependent on temple or court. But naturally priests, civil and military leaders and officials absorbed a major share of the concentrated surplus and thus formed a 'ruling class.' Unlike a palaeolithic magician or a neolithic chief, they were, as an Egyptian scribe actually put it, 'exempt from all manual tasks.' On the other hand, the lower classes were not only guaranteed peace and security, but were relieved from intellectual tasks which many find more irksome than any physical labour. Besides reassuring the masses that the sun was going to rise next day and the river would flood again next year (people who have not five thousand years of recorded experience of natural uniformities behind them are really worried about such matters!), the ruling classes did confer substantial benefits upon their subjects in the way of planning and organization.

(6) They were in fact compelled to invent systems of recording and exact, but practically useful, sciences.

The mere administration of the vast revenues of a Sumerian temple or an Egyptian pharaoh by a perpetual corporation of priests or officials obliged its members to devise conventional methods of recording that should be intelligible to all their colleagues and successors, that is, to invent systems of writing and numeral notation. Writing is thus a significant, as well as a convenient, mark of civilization. But while writing is a trait common to Egypt, Mesopotamia, the Indus valley and Central America, the characters themselves were different in each region and so were the normal writing materials – papyrus in Egypt, clay in Mesopotamia. The engraved seals or stelae that provide the sole extant evidence for early Indus and Maya writing no more represent the normal vehicles for the scripts than do the comparable documents from Egypt and Sumer.

(7) The invention of writing – or shall we say the inventions of scripts – enabled the leisured clerks to proceed to the elaboration of exact and predictive sciences – arithmetic, geometry and astronomy. Obviously beneficial and explicitly attested by the Egyptian and Maya documents was the correct determination of the tropic year and the creation of a calendar. For it enabled the rulers to regulate successfully the cycle of agricultural operations. But once more the Egyptian, Maya and Babylonian calendars were as different as any systems based on a single natural unit could be. Calendrical and mathematical sciences are common features of the earliest civilizations and they too are corollaries of the archaeologists' criterion, writing.

(8) Other specialists, supported by the concentrated social surplus, gave a new direction to artistic expression. Savages even in palaeolithic times had tried, sometimes with astonishing success, to depict animals and even men as they saw them – concretely and naturalistically. Neolithic peasants never did that; they hardly ever tried to represent natural objects, but preferred to symbolize them by abstract geometrical patterns which at most may suggest by a few traits a fantastical man or beast or plant. But Egyptian, Sumerian, Indus and Maya artist-craftsmen – full-time sculptors, painters, or seal-engravers – began once more to carve, model or draw likenesses of persons or things, but no longer with the naive naturalism of the hunter, but according to conceptualized and sophisticated styles which differ in each of the four urban centres.

(9) A further part of the concentrated social surplus was used to pay for the importation of raw materials, needed for industry or cult and not available locally. Regular 'foreign' trade over quite long distances was a feature of all early civilizations and, though common enough among barbarians later, is not certainly attested in the Old World before 3000 B.C. nor in the New before the Maya 'empire.' Thereafter regular trade extended from Egypt at least as far as Byblos on the Syrian coast while Mesopotamia was related by commerce with the Indus valley. While the objects of international trade were at first mainly 'luxuries,' they already included industrial materials, in the Old World notably metal, the place of which in the New was perhaps taken by obsidian. To this extent the first cities were dependent for vital materials on long distance trade as no neolithic village ever was.

(10) So in the city, specialist craftsmen were both provided with raw materials needed for the employment of their skill and also guaranteed security in a State organization based now on residence rather than kinship. Itinerancy was no longer obligatory. The city was a community to which a craftsman could belong politically as well as economically.

Yet in return for security they became dependent on temple or court and were relegated to the lower classes. The peasant masses gained even less material advantages; in Egypt for instance metal did not replace the old stone and wood tools for agricultural work. Yet, however imperfectly, even the earliest urban communities must have been held together by a sort of solidarity missing from any neolithic village. Peasants, craftsmen, priests and rulers form a community, not only by reason of identity of language and belief, but also because each performs mutually complementary functions, needed for the well-being (as redefined under civilization) of the whole. In fact the earliest cities illustrate a first approximation to an organic solidarity based upon a functional complementarity and interdependence between all its members such as subsist between the constituent cells of an organism. Of course this was only a very distant approximation. However necessary the concentration of the surplus really was with the existing forces of production, there seemed a glaring conflict on economic interests between the tiny ruling class, who annexed the bulk of the social surplus, and the vast majority who were left with a bare subsistence and effectively excluded from the spiritual benefits of civilization. So solidarity had still to be maintained by the ideological devices appropriate to the mechanical solidarity of barbarism as expressed in the pre-eminence of the temple or the sepulchral shrine, and now supplemented by the force of the new State organization. There could be no room for sceptics or sectaries in the oldest cities.

These ten traits exhaust the factors common to the oldest cities that archaeology, at best helped out with fragmentary and often ambiguous written sources, can detect. No specific elements of town planning for example can be proved characteristic of all such cities; for on the one hand the Egyptian and Maya cities have not yet been excavated; on the other neolithic villages were often walled, an elaborate system of sewers drained the Orcadian hamlet of Skara Brae; two-storeyed houses were built in pre-Columbian pueblos, and so on.

The common factors are quite abstract. Concretely Egyptian, Sumerian, Indus and Maya civilizations were as different as the plans of their temples, the signs of their scripts and their artistic conventions. In view of this divergence and because there is so far no evidence for a temporal priority of one Old World centre (for instance, Egypt) over the rest nor yet for contact between Central America and any other urban centre, the four revolutions just considered may be regarded as mutually independent. On the contrary, all later civilizations in the Old World may in a sense be regarded as lineal descendants of those of Egypt, Mesopotamia or the Indus.

But this was not a case of like producing like. The maritime civilizations of Bronze Age Crete or classical Greece for example, to say nothing of our own, differ more from their reputed ancestors than these did among themselves. But the urban revolutions that gave them birth did not start from scratch. They could and probably did draw upon the capital accumulated in the three allegedly primary centres. That is most obvious in the case of cultural capital. Even today we use the Egyptians' calendar and the Sumerians' divisions of the day and the hour. Our European ancestors did not have to invent for themselves these divisions of time nor repeat the observations on which they are based; they took over – and very slightly improved – systems elaborated 5,000 years ago! But the same is in a sense true of material capital as well. The Egyptians, the Sumerians and the Indus people had accumulated vast reserves of surplus food. At the same time they had to import from abroad necessary raw materials like metals and building timber as well as 'luxuries.' Communities controlling these natural resources could in exchange claim a slice of the urban surplus. They could use it as capital to support full-time specialists – craftsmen or rulers – until the latters' achievement in technique and organization had so enriched barbarian economics that they too could produce a substantial surplus in their turn.

“The Urbanization of the Human Population”

Scientific American (1965)

Kingsley Davis

EDITORS’ INTRODUCTION

The demographics of the urbanization process are the foundation of all urban history. Demography – from the Greek *demos*: “people” – is the study of human populations. Kingsley Davis (1908–1996) pioneered the study of historical urban demography and was particularly fascinated by the history of world urbanization: that is, the increase over time of the proportion of the total human population that is urban as opposed to rural.

The following selection synthesizes Davis’s conclusions about how urbanization has occurred throughout the world during all of human history. He raises fundamental issues and lays out a clear framework for understanding population dynamics and urban growth. Davis’s careful distinctions of possible sources of urbanization are fundamental. He concludes that, historically, urbanization is primarily caused by rural–urban migration, not because of other possible factors such as differential birth and mortality rates.

Davis’s extraordinary data on how tiny European urban settlements were after the fall of Rome, and how slowly they grew throughout the Middle Ages and early modern period, provides the demographic backdrop for the historic growth of European urbanization. During the long period of medieval urbanization, the proportion of the population that was urban as opposed to rural changed very slowly. In sharp contrast, urbanization increases very rapidly around the year 1800, and Davis concludes that as the Industrial Revolution in England, along with rapid population growth, combined with rural–urban shifts to change both the proportion of the population living in cities and absolute city size very quickly. Friedrich Engels (p. 53) describes in horrifying detail what this revolution in urban demography meant to the impoverished urban proletariat of Manchester and other nineteenth-century industrial cities. His analysis is extremely relevant in assessing prospects for the twenty-first century as the advanced industrial societies and eventually the world reach what some environmental analysts regard as the full “carrying capacity” of the globe.

Davis argues that urbanization follows an attenuated S-curve in which pre-industrial cities urbanize very slowly at the long bottom of the S, shoot up at the middle of the S as they industrialize, and then level off at the top of the S (see Plate 1). He observes that advanced industrialized countries are now reaching the top part of an S-curve, many rapidly urbanizing Third World countries are at the steep middle of the S, and other emerging countries are still moving along the long slowly rising bottom of the S. Davis describes a “family of S curves” on a horizontal x-axis representing time and a vertical y-axis representing the percentage of a country’s population that is urban. England (which was only about 5 percent urban in 1300 CE, but is now about 93 percent urban) is at the top of the family of S-curves. The steep part of England’s S starts about 1750 with the start of the Industrial Revolution. A long, nearly level top to the S shows that England became nearly fully urbanized several decades ago. The long attenuated parts of the S-curves for Germany and France begin later and the time at which the S starts to rise rapidly also occurs later than in England. The long bottom part of China’s S-curve extends until about 1980 and only then begins to rise rapidly. The S-curves for a few very poor African countries still have only the bottom part of the S because they have not yet begun to urbanize rapidly.

The developing countries of Asia, South America, and Africa already have many huge and rapidly growing cities. As the twenty-first century progresses, it appears likely the human population will increasingly live in “megacities” of ten million inhabitants and more, often flowing together in vast urban conurbations sometimes called “mega-urban regions.”

Davis concludes that there will be an end to urbanization – but not necessarily to absolute population growth, the physical size of cities, or the absolute number of people cities contain. He found that the rural population in “Third World” or underdeveloped countries today continues to grow as these countries urbanize, unlike European cities in the nineteenth century where industrialization led to depopulation of rural areas. His vision of developing societies unable to sustain their populations helps to explain Saskia Sassen’s description of growing poverty and inequality worldwide and the growth of large, poorly paid immigrant labor forces in the largest cities in the developed world (p. 650). Research and scholarly debate continues on the nature and causes of world urbanization. The most authoritative measures of world population and urbanization growth come from the Population Division of the Department of Economic and Social Affairs of the United Nations. According to that organization’s *World Population to 2300* (New York: United Nations, 2004), the total world population is expected to peak at 9.22 billion in 2075 and then level off and slowly decline “to reach a level of 8.97 billion by 2300.” The same organization’s *World Urbanization Prospects, the 2011 Revision* (New York: United Nations, 2012), projects that between “2011 and 2050, the world population is expected to increase by 2.3 billion, passing from 7.0 billion to 9.3 billion. At the same time, the population living in urban areas is projected to gain 2.6 billion, passing from 3.6 billion in 2011 to 6.3 billion in 2050.” And in 2014, a team of demographers reviewed the data in the UN 2011 revision and predicted that world population might well peak at as much as 11 billion, based on the continuing high birth rates in Africa. Only time will tell which prediction will prove to be accurate, but it is clear that the increase in urbanization is worldwide and extraordinary. The world is already more than 50 percent urban – that benchmark was passed sometime in 2007–2008 – and by 2020 “it is expected that half of the population of Asia will live in urban areas, while Africa is likely to reach a 50 percent urbanization rate . . . in 2035.” Overall, world urbanization is expected to increase to 72 percent by the year 2050.

Historians continue to shed light on the growth of cities, but because the records from which they work are often fragmentary and incomplete not everyone agrees with Davis or any other standard account. Debate continues on the relative importance of war, plague, medical advances, trade, technology, religion, and ideology on urban growth. And debate is even more intense in the normative area – about what, if anything, governments should do about population growth and urbanization. Davis stressed the impact of overall population growth (which he saw as a real danger) on world urbanization and implies that family planning is essential if cities are to meet human needs. But many governments reject family planning on religious or policy grounds, and some European countries now face declining populations and are currently debating the desirability of enacting family-friendly policies to reward childbearing. And today a whole new dimension has been added to the urbanization debate involving the very definition of what is urban, what is a city. As Manuel Castells (p. 229) and others have argued, many urban functions are now being carried out online, in the electronic “space of flows,” meaning that some forms of urban life can be carried out in suburban, even ex-urban physical spaces. In the 1960s, Canadian cultural critic Marshall McLuhan argued that the new television medium would result in a world that had become a “global village.” Perhaps the even newer digital telecommunications media will allow humanity to reach an effective 100 percent urbanization rate in a world that has become a single “global city” (see Saskia Sassen, p. 650) or a “global city network” (see Peter J. Taylor, p. 92).

Kingsley Davis’s other writings include many articles and studies on demographics and natural resources as well as two anthologies: *Cities: Their Origin, Growth and Human Impact* (San Francisco: W.H. Freeman, 1973) and, with Mikhail S. Bernstram, *Resources, Environment and Population: Present Knowledge, Future Options* (New York: Population Council, Oxford University Press, 1991). For more on Davis and his writings, see David Hoer, *Kingsley Davis: A Biography and Selections from His Writings* (Edison, NJ: Transaction Publishers, 2004).

Important data on world urbanization are contained in Tertius Chandler and Gerald Fox, *3000 Years of Urban Growth* (New York: Academic Press, 1974) and Tertius Chandler, *Four Thousand Years of Urban Growth: An*

Historical Census (Lewiston, NY: Edwin Mellen Press, 1987). These books assemble estimates of the population size of individual cities everywhere in the world over four millennia. The sources of the estimates range from contemporary accounts to scholarly estimates completed just before the second of the two books was published in 1987. Footnotes explain where the estimates come from. This is a valuable compendium for source information. The authors attempt to synthesize the source material so that they provide longitudinal data on the population of different cities over time, however, are problematic. Since the sources are so varied and conflicting and not based on consistent definitions or methods Chandler and Fox's estimates – particularly for the earliest time periods and for cities where the records are least complete and reliable – must be judged with extreme caution. Much more reliable data on the population of all Western European cities that achieved a population of 10,000 or more at any time between 1500 and 1800 are reported at fifty-year increments starting in 1500 in Jan DeVries, *European Urbanization, 1500–1800* (Cambridge, MA: Harvard University Press, 1984). Further insight on demography and urbanization can be found in World Bank, *World Development Indicators, 2012* (Washington: World Bank, 2012), Ad van der Woude, Akira Hayami, and Jan de Vries (eds.), *Urbanization in History: A Process of Dynamic Interaction* (Oxford: Oxford University Press, 1990), and the frequent revisions of *World Population Prospects* published by the Population Division of the United Nations. Also useful is Paul Knox and Linda McCarthy, *Urbanization: An Introduction to Urban Geography*, 2nd edn (Upper Saddle River, NJ: Prentice-Hall, 2011).

For recent developments in urbanization in the underdeveloped nations of the world, see Alan Gilbert (ed.), *The Mega-City in Latin America* (New York: United Nations University Press, 1996), Carole Rakodi (ed.), *The Urban Challenge in Africa* (New York: United Nations University Press, 1997), and Fu-chen Lo and Yue-man Yeung (eds.), *Emerging World Cities in Pacific Asia* (New York: United Nations University Press, 1996).

For an environmental view of world urbanization, consult Cedric Pugh (ed.), *Sustainability, the Environment, and Urbanization* (London: Earthscan, 1996). George Martine et al., *The New Global Frontier: Urbanization, Poverty and Environment in the 21st Century* (London: Earthscan, 2008) provides useful data and analysis about recent world urbanization. The future of urbanization is of course an important issue for policy planners. For a fascinating review of population-related policy issues, including the possibility of a "world population implosion," see Nicholas Eberstadt, *Prosperous Paupers and Other Population Problems* (New Brunswick, NJ: Transaction, 2000). For more on the possibility of declining populations in the future, consult Phillip Longman, *The Empty Cradle: How Falling Birthrates Threaten World Prosperity and What To Do About It* (New York: Basic Books, 2004), Ben J. Wattenburg, *Fewer: How the Demography of Depopulation Will Shape Our Future* (New York: Ivan R. Dee, 2004), and, for the special case of the effects of China's "one child" policy, Valerie Hudson, *Bare Essentials: The Security Implications of Asia's Surplus Male Population* (Cambridge, MA: MIT Press, 2004).

Urbanized societies, in which a majority of the people live crowded together in towns and cities, represent a new and fundamental step in man's social evolution. Although cities themselves first appeared some 5,500 years ago, they were small and surrounded by an overwhelming majority of rural people; moreover, they relapsed easily to village or small-town status. The urbanized societies of today, in contrast, not only have urban agglomerations of a size never before attained but also have a high proportion of their population concentrated in such agglomerations. In 1960, for example, nearly 52 million Americans lived in only 16 urbanized areas. Together these areas covered less land than one of the smaller counties (Cochise) of Arizona. According to one definition used by the U.S. Bureau of the Census, 96 million people – 53 percent of the nation's population – were concentrated in 213 urbanized areas that together occupied only 0.7 percent of the nation's land. Another definition used by the bureau puts the urban population at about 70 percent. The large and dense agglomerations comprising the urban population involve a degree of human contact and of social complexity never before known. They exceed in size the communities of any other large animal; they suggest the behavior of communal insects rather than of mammals.

Neither the recency nor the speed of this evolutionary development is widely appreciated. Before 1850 no society could be described as predominantly urbanized, and by 1900 only one – Great Britain – could be so regarded. Today, only 65 years later, all industrial nations are highly urbanized, and in the world as a whole the process of urbanization is accelerating rapidly.

Some years ago my associates and I at Columbia University undertook to document the progress of urbanization by compiling data on the world's cities and the proportion of human beings living in them; in recent years the work has been continued in our center – International Population and Urban Research – at the University of California at Berkeley. The data obtained in these investigations . . . show the historical trend in terms of one index of urbanization: the proportion of the population living in cities of 100,000 or larger. Statistics of this kind are only approximations of reality, but they are accurate enough to demonstrate how urbanization has accelerated. Between 1850 and 1950 the index changed at a much higher rate than from 1800 to 1850, but the rate of change from 1950 to 1960 was twice that of the preceding 50 years! If the pace of increase that obtained between 1950 and 1960 were to remain the same, by 1990 the fraction of the world's people living in cities of 100,000 or larger would be more than half. Using another index of urbanization – the proportion of the world's population living in urban places of all sizes – we found that by 1960 the figure had already reached 33 percent.

Clearly the world as a whole is not fully urbanized, but it soon will be. This change in human life is so recent that even the most urbanized countries still exhibit the rural origins of their institutions. Its full implications for man's organic and social evolution can only be surmised.

In discussing the trend – and its implications insofar as they can be perceived – I shall use the term "urbanization" in a particular way. It refers here to the proportion of the total population concentrated in urban settlements, or else to a rise in this proportion. A common mistake is to think of urbanization as simply the growth of cities. Since the total population is composed of both the urban population and the rural, however, the "proportion urban" is a function of both of them. Accordingly, cities can grow without any urbanization, provided that the rural population grows at an equal or a greater rate.

Historically, urbanization and the growth of cities have occurred together, which accounts for the confusion. As the reader will soon see, it is necessary to distinguish the two trends. In the most advanced countries today, for example, urban populations are still growing, but their proportion of the total population is tending to remain stable or to diminish. In other words, the process of urbanization – the switch from a spread-out pattern of human settlement to one of concentration in urban centers – is a change that has a beginning and an end, but the growth of cities has no inherent limit. Such growth could continue even after everyone was living in cities, through sheer excess of births over deaths.

The difference between a rural village and an urban community is of course one of degree; a precise operational distinction is somewhat arbitrary, and it varies from one nation to another. Since data are available for communities of various sizes, a dividing line can be chosen at will. One convenient index of urbanization, for example, is the proportion of people living in places of 100,000 or more. In the following analysis I shall depend on two indexes: the one just mentioned and the proportion of population classed as "urban" in the official statistics of each country. In practice the two indexes are highly correlated; therefore either one can be used as an index of urbanization.

Actually the hardest problem is not that of determining the "floor" of the urban category but of ascertaining the boundary of places that are clearly urban by any definition. How far east is the boundary of Los Angeles? Where along the Hooghly River does Calcutta leave off and the countryside begin? In the past the population of cities and towns has usually been given as the number of people living within the political boundaries. Thus the population of New York is frequently given as around eight million, this being the population of the city proper. The error in such a figure was not large before World War I, but since then, particularly in the advanced countries, urban populations have been spilling over the narrow political boundaries at a tremendous rate. In 1960 the New York–Northeastern New Jersey urbanized area, as delineated by the Bureau of the Census, had more than 14 million people. That delineation showed it to be the largest city in the world and nearly twice as large as New York City proper.

As a result of the outward spread of urbanites, counts made on the basis of political boundaries alone underestimate the city populations and exaggerate the rural. For this reason our office delineated the metropolitan areas of as many countries as possible for dates around 1950. These areas included the central, or political, cities and the zones around them that are receiving the spillover.

This reassessment raised the estimated proportion of the world's population in cities of 100,000 or larger from 15.1 percent to 16.7 percent. As of 1960 we have used wherever possible the "urban agglomeration" data now furnished to the United Nations by many countries. The U.S., for example, provides data for "urbanized areas," meaning cities of 50,000 or larger and the built-up agglomerations around them.

. . . My concern is with the degree of urbanization in whole societies. It is curious that thousands of years elapsed between the first appearance of small cities and the emergence of urbanized societies in the nineteenth century. It is also curious that the region where urbanized societies arose – northwestern Europe – was not the one that had given rise to the major cities of the past; on the contrary, it was a region where urbanization had been at an extremely low ebb. Indeed, the societies of northwestern Europe in medieval times were so rural that it is hard for modern minds to comprehend them. Perhaps it was the nonurban character of these societies that erased the parasitic nature of towns and eventually provided a new basis for a revolutionary degree of urbanization.

At any rate, two seemingly adverse conditions may have presaged the age to come: one the low productivity of medieval agriculture in both per-acre and per-man terms, the other the feudal social system. The first meant that towns could not prosper on the basis of local agriculture alone but had to trade and to manufacture something to trade. The second meant that they could not gain political dominance over their hinterlands and thus become warring city-states. Hence they specialized in commerce and manufacture and evolved local institutions suited to this role. Craftsmen were housed in the towns, because there the merchants could regulate quality and cost. Competition among towns stimulated specialization and technological innovation. The need for literacy, accounting skills and geographical knowledge caused the towns to invest in secular education.

Although the medieval towns remained small and never embraced more than a minor fraction of each region's population, the close connection between industry and commerce that they fostered, together with their emphasis on technique, set the stage for the ultimate breakthrough in urbanization. This breakthrough came only with the enormous growth in productivity caused by the use of inanimate energy and machinery. How difficult it was to achieve the transition is agonizingly apparent from statistics showing that even with the conquest of the New World the growth of urbanization during three postmedieval centuries in Europe was barely perceptible. I have assembled population estimates at two or more dates for 33 towns and cities in the sixteenth century, 46 in the seventeenth and 61 in the eighteenth. The average rate of growth during the three centuries was less than 0.6 percent per year. Estimates of the growth of Europe's population as a whole between 1650 and 1800 work out to slightly more than 0.4 percent. The advantage of the towns was evidently very slight. Taking only the cities of 100,000 or more inhabitants, one finds that in 1600 their combined population was 1.6 percent of the estimated population of Europe; in 1700, 1.9 percent; and in 1800, 2.2 percent. On the eve of the industrial revolution Europe was still an overwhelmingly agrarian region.

With industrialization, however, the transformation was striking. By 1801 nearly a tenth of the people of England and Wales were living in cities of 100,000 or larger. This proportion doubled in 40 years and doubled again in another 60 years. By 1900 Britain was an urbanized society. In general, the later each country became industrialized, the faster was its urbanization. The change from a population with 10 percent of its members in cities of 100,000 or larger to one in which 30 percent lived in such cities took about 79 years in England and Wales, 66 in the U.S., 48 in Germany, 36 in Japan and 26 in Australia. The close association between economic development and urbanization has persisted: . . . in 199 countries around 1960 the proportion of the population living in cities varied sharply with per capita income.

Clearly, modern urbanization is best understood in terms of its connection with economic growth, and its implications are best perceived in its latest manifestations in advanced countries. What becomes apparent as one examines the trend in these countries is that urbanization is a finite process, a cycle through which nations go in their transition from agrarian to industrial society. The intensive urbanization of most of the advanced countries began within the past hundred years; in the underdeveloped countries it got under way more recently. In some of the advanced countries its end is now in sight. The fact that it will end, however, does not mean that either economic development or the growth of cities will necessarily end.

The typical cycle of urbanization can be represented by a curve in the shape of an attenuated S. Starting from the bottom of the S, the first bend tends to come early and to be followed by a long attenuation. In the United Kingdom, for instance, the swiftest rise in the proportion of people living in cities of 100,000 or larger occurred from 1811 to 1851. In the U.S. it occurred from 1820 to 1890, in Greece from 1879 to 1921. As the proportion climbs above 50 percent the curve begins to flatten out; it falters, or even declines, when the proportion urban has reached about 75 percent. In the United Kingdom, one of the world's most urban countries, the proportion was slightly higher in 1926 (78.7 percent) than in 1961 (78.3 percent).

At the end of the curve some ambiguity appears. As a society becomes advanced enough to be highly urbanized it can also afford considerable suburbanization and fringe development. In a sense the slowing down of urbanization is thus more apparent than real: an increasing proportion of urbanites simply live in the country and are classified as rural. Many countries now try to compensate for this ambiguity by enlarging the boundaries of urban places; they did so in numerous censuses taken around 1960. Whether in these cases the old classification of urban or the new one is erroneous depends on how one looks at it; at a very advanced stage the entire concept of urbanization becomes ambiguous.

The end of urbanization cannot be unraveled without going into the ways in which economic development governs urbanization. Here the first question is: where do the urbanites come from? The possible answers are few: the proportion of people in cities can rise because rural settlements grow larger and are reclassified as towns or cities; because the excess of births over deaths is greater in the city than in the country, or because people move from the country to the city.

The first factor has usually had only slight influence. The second has apparently never been the case. Indeed, a chief obstacle to the growth of cities in the past has been their excessive mortality. London's water in the middle of the nineteenth century came mainly from wells and rivers that drained cesspools, graveyards and tidal areas. The city was regularly ravaged by cholera. Tables for 1841 show an expectation of life of about 36 years for London and 26 for Liverpool and Manchester, as compared to 41 for England and Wales as a whole. After 1850, mainly as a result of sanitary measures and some improvement in nutrition and housing, city health improved, but as late as the period 1901–1910 the death rate of the urban counties in England and Wales, as modified to make the age structure comparable, was 33 percent higher than the death rate of the rural counties. As Bernard Benjamin, a chief statistician of the British General Register Office, has remarked: "Living in the town involved not only a higher risk of epidemic and crowd diseases . . . but also a higher risk of degenerative disease – the harder wear and tear of factory employment and urban discomfort." By 1950, however, virtually the entire differential had been wiped out.

As for birth rates, during rapid urbanization in the past they were notably lower in cities than in rural areas. In fact, the gap tended to widen somewhat as urbanization proceeded in the latter half of the nineteenth century and the first quarter of the twentieth. In 1800 urban women in the U.S. had 36 percent fewer children than rural women did; in 1840, 38 percent and in 1930, 41 percent. Thereafter the difference diminished.

With mortality in the cities higher and birth rates lower, and with reclassification a minor factor, the only real source for the growth in the proportion of people in urban areas during the industrial transition was rural–urban migration. This source had to be plentiful enough not only to overcome the substantial disadvantage of the cities in natural increase but also, above that, to furnish a big margin of growth in their populations. If, for example, the cities had a death rate a third higher and a birth rate a third lower than the rural rates (as was typical in the latter half of the nineteenth century), they would require each year perhaps 40 to 45 migrants from elsewhere per 1,000 of their population to maintain a growth rate of 3 percent per year. Such a rate of migration could easily be maintained as long as the rural portion of the population was large, but when this condition ceased to obtain, the maintenance of the same urban rate meant an increasing drain on the countryside.

Why did the rural–urban migration occur? The reason was that the rise in technological enhancement of human productivity, together with certain constant factors, rewarded urban concentration. One of the constant factors was that agriculture uses land as its prime instrument of production and hence spreads out people who are engaged in it, whereas manufacturing, commerce and services use land only as a site. Moreover, the demand for agricultural products is less elastic than the demand for services and manufactures. As productivity grows, services and manufactures can absorb more manpower by paying higher wages. Since nonagricultural activities can use land simply as a site, they can locate near one another (in towns and cities) and thus minimize the fraction of space inevitably involved in the division of labor. At the same time, as agricultural technology is improved, capital costs in farming rise and manpower becomes not only less needed but also economically more burdensome. A substantial portion of the agricultural population is therefore sufficiently disadvantaged, in relative terms, to be attracted by higher wages in other sectors.

In this light one sees why a large *flow* of people from farms to cities was generated in every country that passed through the industrial revolution. One also sees why, with an even higher proportion of people already in cities and with the inability of city people to replace themselves by reproduction, the drain eventually became so heavy that in many nations the rural population began to decline in absolute as well as relative terms. In Sweden it declined after 1920, in England and Wales after 1861, in Belgium after 1910.

Realizing that urbanization is transitional and finite, one comes on another fact – a fact that throws light on the circumstances in which urbanization comes to an end. A basic feature of the transition is the profound switch from agricultural to nonagricultural employment. This change is associated with urbanization but not identical with it. The difference emerges particularly in the later stages. Then the availability of automobiles, radios, motion pictures and electricity, as well as the reduction of the workweek and the workday, mitigate the disadvantages of living in the country. Concurrently the expanding size of cities makes them more difficult to live in. The population classed as "rural" is accordingly enlarged, both from cities and from true farms. For these reasons the "rural" population in some industrial countries never did fall in absolute size. In all the industrial countries, however, the population dependent on agriculture – which the reader will recognize as a more functional definition of the nonurban population than mere rural residence – decreased in absolute as well as relative terms. In the U.S., for example, the net migration from farms totaled more than 27 million between 1920 and 1959 and thus averaged approximately 700,000 a year. As a result the farm population declined from 32.5 million in 1916 to 20.5 million in 1960, in spite of the large excess of births in farm families. In 1964, by a stricter American definition classifying as "farm families" only those families actually earning their living from agriculture, the farm population was down to 12.9 million. This number represented 6.8 percent of the nation's population; the comparable figure for 1880 was 44 percent. In Great Britain the number of males occupied in agriculture was, at its peak, 1.8 million, in 1851; by 1961 it had fallen to 0.5 million.

In the later stages of the cycle, then, urbanization in the industrial countries tends to cease. Hence the connection between economic development and the growth of cities also ceases. The change is explained by two circumstances. First, there is no longer enough farm population to furnish a significant migration to the cities. (What can 12.9 million American farmers contribute to the growth of the 100 million people already in urbanized areas?) Second, the rural nonfarm population, nourished by refugees from the expanding cities, begins to increase as fast as the city population. The effort of census bureaus to count fringe residents as urban simply pushes the definition of "urban" away from the notion of dense settlement and in the direction of the term "nonfarm." As the urban population becomes more "rural," which is to say less densely settled, the advanced industrial peoples are for a time able to enjoy the amenities of urban life without the excessive crowding of the past.

Here, however, one again encounters the fact that a cessation of urbanization does not necessarily mean a cessation of city growth. An example is provided by New Zealand. Between 1945 and 1961 the proportion of New Zealand's population classed as urban – that is, the ratio between urban and rural residents – changed hardly at all (from 61.3 percent to 63.6 percent) but the urban population increased by 50 percent. In Japan between 1940 and 1950 urbanization actually decreased slightly, but the urban population increased by 13 percent.

The point to be kept in mind is that once urbanization ceases, city growth becomes a function of general population growth. Enough farm-to-city migration may still occur to redress the difference in natural increase. The reproductive rate of urbanites tends, however, to increase when they live at lower densities, and the reproductive rate of "urbanized" farmers tends to decrease; hence little migration is required to make the urban increase equal the national increase.

I now turn to the currently underdeveloped countries. With the advanced nations having slackened their rate of urbanization, it is the others – representing three-fourths of humanity – that are mainly responsible for the rapid urbanization now characterizing the world as a whole. In fact, between 1950 and 1960 the proportion of the population in cities of 100,000 or more rose about a third faster in the underdeveloped regions than in the developed ones. Among the underdeveloped regions the pace was slow in eastern and southern Europe but in the rest of the underdeveloped world the proportion in cities rose twice as fast as it did in the industrialized countries, even though the latter countries in many cases broadened their definitions of urban places to include more suburban and fringe residents.

Because of the characteristic pattern of urbanization, the current rates of urbanization in underdeveloped countries could be expected to exceed those now existing in countries far advanced in the cycle. On discovering that this is the case one is tempted to say that the underdeveloped regions are now in the typical stage of urbanization associated with early economic development. This notion, however, is erroneous. In their urbanization the underdeveloped countries are definitely not recreating past history. Indeed, the best grasp of their present situation comes from analyzing how their course differs from the previous pattern of development.

The first thing to note is that today's underdeveloped countries are urbanizing not only more rapidly than the industrial nations are now but also more rapidly than the industrial nations did in the heyday of their urban growth. The difference, however, is not large. In 40 underdeveloped countries for which we have data in recent decades, the average gain in the proportion of the population urban was 20 percent per decade; in 16 industrial countries, during the decades of their most rapid urbanization (mainly in the nineteenth century), the average gain per decade was 15 percent.

This finding that urbanization is proceeding only a little faster in underdeveloped countries than it did historically in the advanced nations may be questioned by the reader. It seemingly belies the widespread impression that cities throughout the nonindustrial parts of the world are bursting with people. There is, however, no contradiction. One must recall the basic distinction between a change in the proportion of the urban population, which is a ratio, and the absolute growth of cities. The popular impression is correct: the cities in underdeveloped areas are growing at a disconcerting rate. They are far outstripping the city boom of the industrializing era in the nineteenth century. If they continue their recent rate of growth, they will double their population every 15 years.

In 34 underdeveloped countries for which we have data relating to the 1940s and 1950s, the average annual gain in the urban population was 4.5 percent. The figure is remarkably similar for the various regions: 4.7 percent in seven countries of Africa, 4.7 percent in 15 countries of Asia and 4.3 percent in 12 countries of Latin America. In contrast, in nine European countries during their period of fastest urban population growth (mostly in the latter half of the nineteenth century) the average gain per year was 2.1 percent. Even the frontier industrial countries – the U.S., Australia, New Zealand, Canada and Argentina – which received huge numbers of immigrants had a smaller population growth in towns and cities: 4.2 percent per year. In Japan and the U.S.S.R. the rate was respectively 5.4 and 4.3 percent per year, but their economic growth began only recently.

How is it possible that the contrast in growth between today's underdeveloped countries and yesterday's industrializing countries is sharper with respect to the absolute urban population than with respect to the urban share of the total population? The answer lies in another profound difference between the two sets of countries – a difference in total population growth, rural as well as urban. Contemporary underdeveloped populations have been growing since 1940 more than twice as fast as industrialized populations, and their increase far exceeds the growth of the latter at the peak of their expansion. The only rivals in an earlier day were the frontier nations, which had the help of great streams of immigrants. Today the underdeveloped nations – already densely settled, tragically impoverished and with gloomy economic prospects – are multiplying their people by sheer biological increase at a rate that is unprecedented. It is this population boom that is overwhelmingly responsible for the rapid inflation of city populations in such countries. Contrary to popular opinion both inside and outside those countries, the main factor is not rural–urban migration.

This point can be demonstrated easily by a calculation that has the effect of eliminating the influence of general population growth on urban growth. The calculation involves assuming that the total population of a given country remained constant over a period of time but that the urban percentage changed as it did historically. In this manner one obtains the growth of the absolute urban population that would have occurred if rural–urban migration were the only factor affecting it. As an example, Costa Rica had in 1927 a total population of 471,500, of which 88,600, or 18.8 percent, was urban. By 1963 the country's total population was 1,325,200 and the urban population was 456,600, or 34.5 percent. If the total population had remained at 471,500 but the urban percentage had still risen from 18.8 to 34.5, the absolute urban population in 1963 would have been only 162,700. That is the growth that would have occurred in the urban population if rural–urban migration had been the only factor. In actuality the urban population rose to 456,600. In other words, only 20 percent of the rapid growth of Costa Rica's towns and cities was attributable to urbanization per se; 44 percent was attributable solely to the country's general population increase, the remainder to the joint operation of both factors. Similarly, in Mexico between 1940 and 1960, 50 percent of the urban population increase was attributable to national multiplication alone and only 22 percent to urbanization alone.

The past performance of the advanced countries presents a sharp contrast. In Switzerland between 1850 and 1888, when the proportion urban resembled that in Costa Rica recently, general population growth alone accounted for only 19 percent of the increase of town and city people, and rural–urban migration alone accounted for 69 percent. In France between 1846 and 1911 only 21 percent of the growth in the absolute urban population was due to general growth alone.

The conclusion to which this contrast points is that one anxiety of governments in the underdeveloped nations is misplaced. Impressed by the mushrooming in their cities of shanty-towns filled with ragged peasants, they attribute the fantastically fast city growth to rural–urban migration. Actually this migration now does little more than make up for the small difference in the birth rate between city and countryside. In the history of the industrial nations, as we have seen, the sizable difference between urban and rural birth rates and death rates required that cities, if they were to grow, had to have an enormous influx of people from farms and villages. Today in the underdeveloped countries the towns and cities have only a slight disadvantage in fertility, and their old disadvantage in mortality not only has been wiped out but also in many cases has been reversed. During the nineteenth century the urbanizing nations were learning how to keep crowded populations in cities from dying like flies. Now the lesson has been learned, and it is being applied to cities even in countries just emerging from tribalism. In fact, a disproportionate share of public health funds goes into cities. As a result, throughout the nonindustrial world people in cities are multiplying as never before, and rural–urban migration is playing a much lesser role.

The trends just described have an important implication for the rural population. Given the explosive overall population growth in underdeveloped countries, it follows that if the rural population is not to pile up on the land and reach an economically absurd density, a high rate of rural–urban migration must be maintained. Indeed, the exodus from rural areas should be higher than in the past. But this high rate of internal movement is not taking place, and there is some doubt that it could conceivably do so.

To elaborate, I shall return to my earlier point that in the evolution of industrialized countries the rural citizenry often declined in absolute as well as relative terms. The rural population of France – 26.8 million in 1846 – was down to 20.8 million by 1926 and 17.2 million by 1962, notwithstanding a gain in the nation's total population during this period. Sweden's rural population dropped from 4.3 million in 1910 to 3.5 million in 1960. Since the category "rural" includes an increasing portion of urbanites living in fringe areas, the historical drop was more drastic and consistent specifically in the farm population. In the U.S., although the "rural" population never quite ceased to grow, the farm contingent began its long descent shortly after the turn of the century; today it is less than two-fifths of what it was in 1910.

This transformation is not occurring in contemporary underdeveloped countries. In spite of the enormous growth of their cities, their rural populations – and their more narrowly defined agricultural populations – are growing at a rate that in many cases exceeds the rise of even the urban population during the evolution of the now advanced countries. The poor countries thus confront a grave dilemma. If they do not substantially step up the exodus from rural areas, these areas will be swamped with underemployed farmers. If they do step up the exodus, the cities will grow at a disastrous rate.

The rapid growth of cities in the advanced countries, painful though it was, had the effect of solving a problem – the problem of the rural population. The growth of cities enabled agricultural holdings to be consolidated, allowed increased capitalization and in general resulted in greater efficiency. Now, however, the underdeveloped countries are experiencing an even more rapid urban growth – and are suffering from urban problems – but urbanization is not solving their rural ills.

A case in point is Venezuela. Its capital, Caracas, jumped from a population of 359,000 in 1941 to 1,507,000 in 1963; other Venezuelan towns and cities equaled or exceeded this growth. Is this rapid rise denuding the countryside of people? No, the Venezuelan farm population increased in the decade 1951–1961 by 11 percent. The only thing that declined was the amount of cultivated land. As a result the agricultural population density became worse. In 1950 there were some 64 males engaged in agriculture per square mile of cultivated land; in 1961 there were 78. (Compare this with 4.8 males occupied in agriculture per square mile of cultivated land in Canada, 6.8 in the U.S. and 15.6 in Argentina.) With each male occupied in agriculture there are of course dependants. Approximately 225 persons in Venezuela are trying to live from each square mile of cultivated land. Most of the growth of cities in Venezuela is attributable to overall population growth. If the general population had not grown at all, and internal migration had been large enough to produce the actual shift in the proportion in cities, the increase in urban population would have been only 28 percent of what it was and the rural population would have been reduced by 57 percent.

The story of Venezuela is being repeated virtually everywhere in the underdeveloped world. It is not only Caracas that has thousands of squatters living in self-constructed junk houses on land that does not belong to them. By whatever name they are called, the squatters are to be found in all major cities in the poorer countries. They live in broad gullies beneath the main plain in San Salvador and on the hillsides of Rio de Janeiro and Bogotá. They tend to occupy with implacable determination parks, school grounds and vacant lots. Amman, the capital of Jordan, grew from 12,000 in 1958 to 247,000 in 1961. A good part of it is slums, and urban amenities are lacking most of the time for most of the people. Greater Baghdad now has an estimated 850,000 people; its slums, like those in many other underdeveloped countries, are in two zones: the central part of the city and the outlying areas. Here are the *sarifa* areas, characterized by self-built reed huts; these areas account for about 45 percent of the housing in the entire city and are devoid of amenities, including even latrines. In addition to such urban problems, all the countries struggling for higher living levels find their rural population growing too and piling up on already crowded land. I have characterized urbanization as a transformation that, unlike economic development, is finally accomplished and comes to an end. At the 1950–1960 rate the term "urbanized world" will be applicable well before the end of the century. One should scarcely expect, however, that mankind will complete its urbanization without major complications. One sign of trouble ahead turns on the distinction I made at the start between urbanization and city growth *per se*. Around the globe today city growth is disproportionate to urbanization. The discrepancy is paradoxical in the industrial nations and worse than paradoxical in the nonindustrial.

It is in this respect that the nonindustrial nations, which still make up the great majority of nations, are far from repeating past history. In the nineteenth and early twentieth centuries the growth of cities arose from and contributed to economic advancement. Cities took surplus manpower from the countryside and put it to work producing goods and services that in turn helped to modernize agriculture. But today in underdeveloped countries, as in present-day advanced nations, city growth has become increasingly un-hinged from economic development and hence from rural–urban migration. It derives in greater degree from overall population growth, and this growth in nonindustrial lands has become unprecedented because of modern health techniques combined with high birth rates.

The speed of world population growth is twice what it was before 1940, and the swiftest increase has shifted from the advanced to the backward nations. In the latter countries, consequently, it is virtually impossible to create city services fast enough to take care of the huge, never-ending cohorts of babies and peasants swelling the urban masses. It is even harder to expand agricultural land and capital fast enough to accommodate the enormous natural increase on farms. The problem is not urbanization, not rural–urban migration, but human multiplication. It is a problem that is new in both its scale and its setting, and runaway city growth is only one of its painful expressions.

As long as the human population expands, cities will expand too, regardless of whether urbanization increases or declines. This means that some individual cities will reach a size that will make nineteenth-century metropolises look like small towns. If the New York urbanized area should continue to grow only as fast as the nation's population (according to medium projections of the latter by the Bureau of the Census), it would reach 21 million by 1985 and 30 million by 2010. I have calculated that if India's population should grow as the U.N. projections indicate it will, the largest city in India in the year 2000 will have between 36 and 66 million inhabitants.

What is the implication of such giant agglomerations for human density? In 1950 the New York–Northeastern New Jersey urbanized area had an average density of 9,810 persons per square mile. With 30 million people in the year 2010, the density would be 24,000 per square mile. Although this level is exceeded now in parts of New York City (which averages about 25,000 per square mile) and many other cities, it is a high density to be spread over such a big area; it would cover, remember, the suburban areas to which people moved to escape high density. Actually, however, the density of the New York urbanized region is dropping, not increasing, as the population grows. The reason is that the territory covered by the urban agglomeration is growing faster than the population: it grew by 51 percent from 1950 to 1960, whereas the population rose by 15 percent.

If, then, one projects the rise in population and the rise in territory for the New York urbanized region one finds the density problem solved. It is not solved for long, though, because New York is not the only city in the region that is expanding. So are Philadelphia, Trenton, Hartford, New Haven and so on. By 1960 a huge stretch of territory about 600 miles long and 30 to 100 miles wide along the eastern seaboard contained some 37 million people. (I am speaking of a longer section of the seaboard than the Boston-to-Washington conurbation referred to by some other authors.) Since the whole area is becoming one big polynucleated city, its population cannot long expand without a rise in density. Thus persistent human multiplication promises to frustrate the ceaseless search for space – for ample residential lots, wide-open suburban school grounds, sprawling shopping centers, one-floor factories, broad freeways.

How people feel about giant agglomerations is best indicated by their headlong effort to escape them. The bigger the city, the higher the cost of space; yet the more the level of living rises, the more people are willing to pay for low-density living. Nevertheless, as urbanized areas expand and collide, it seems probable that life in low-density surroundings will become too dear for the great majority.

One can of course imagine that cities may cease to grow and may even shrink in size while the population in general continues to multiply. Even this dream, however, would not permanently solve the problem of space. It would eventually obliterate the distinction between urban and rural, but at the expense of the rural.

It seems plain that the only way to stop urban crowding and to solve most of the urban problems besetting both the developed and the underdeveloped nations is to reduce the overall rate of population growth. Policies designed to do this have as yet little intelligence and power behind them. Urban planners continue to treat population growth as something to be planned for, not something to be itself planned. Any talk about applying brakes to city growth is therefore purely speculative, overshadowed as it is by the reality of uncontrolled population increase.